# ***Transfer Learning Models***

- *Generally we trained the model on the given data*
- *ML or DL, we save the model in one format*
- *pickle or Job lib For Machine learning*
- *Deep learning tf. save or using keras also*
- *In the deep learning model saving means , saving the weight file*
- *You develop the model with sequential order*
- *This is called model configuration*
- *Model config ==== > Model architecture*
- *Model layers means ==== each layer*
- *Model summary ==== each layer parameters*
- *Model weights means ===== final weights*
- *we developed and we saved: Custom model*
- *I given that saved model to the clinet*
- *Client will use that model : Transfer learning*
- *The models which are useing as Transfer learning called as Pretrained models*
- *Pre trained means already trained model*

*Now a days many companies are training on enemorous data*

*and they are mkaing publicly available model*

*free or pay as you go*

*Cloud AI*

*GenAI*

*CNN: Object detection*

*LeNET ======== Pretrained models*

*AlexNet*

*VGGnet*

*VGG16 – 32*

*MobileNet*

*Google Net*

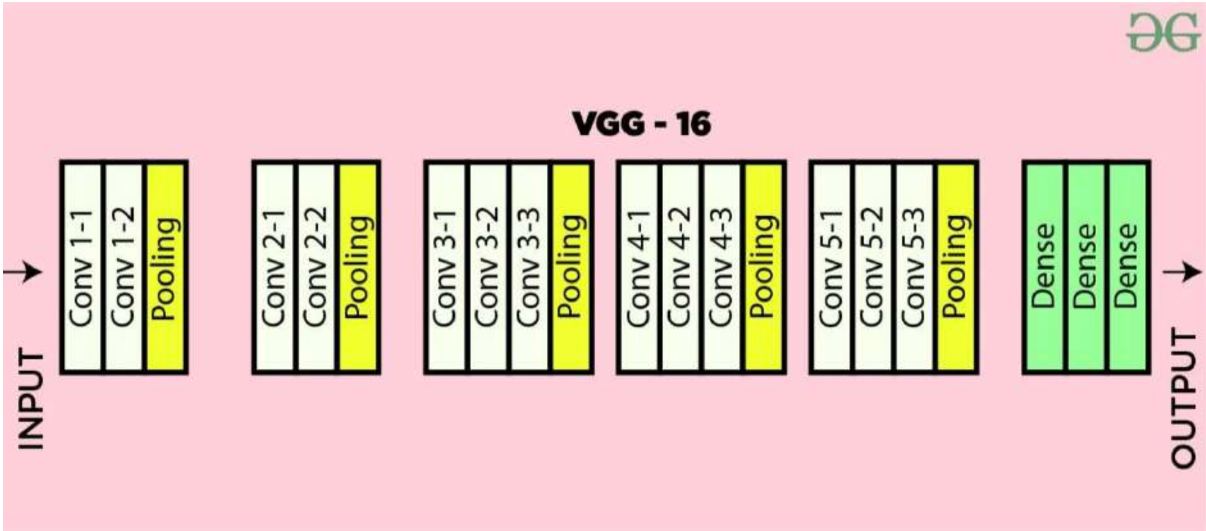
*ResNet*

*FastRCNN*

*MaskRCNN*

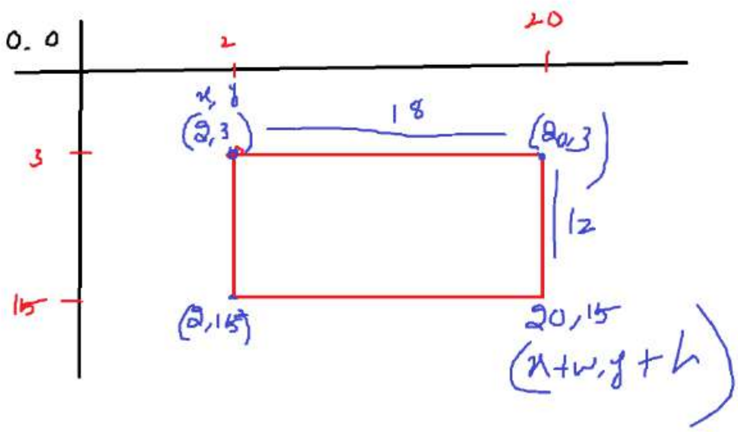
*FasterRCNN*

$YOLO\ V8$

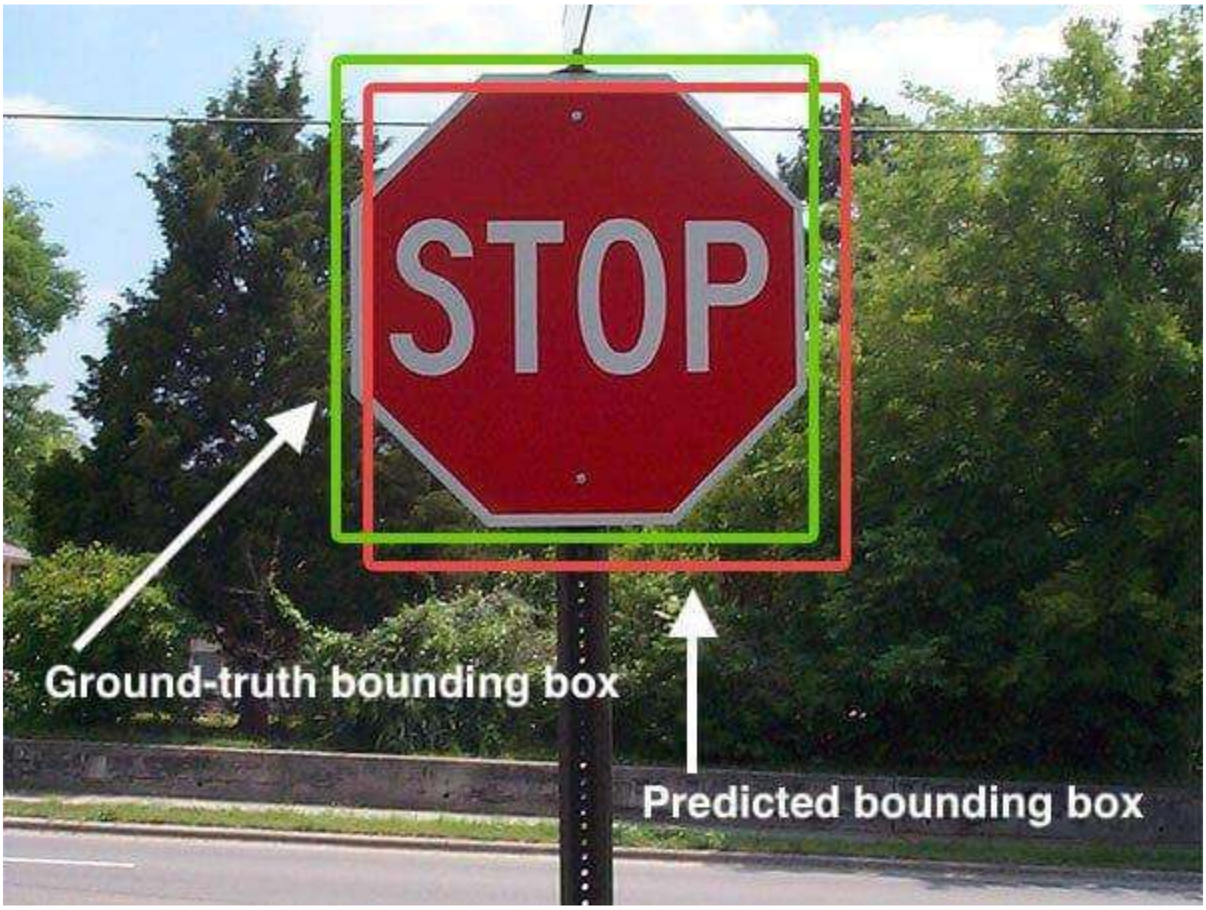



- $Consider\ MobileNet\ \ trained\ on\ 1.3Million\ images$
- $Identify\ the\ objects\ around\ 1000\ classes$
- $Object\ identification\ model$
- $we\ have\ two\ types\ of\ use\ cases$
    - $Object\ identification: recognize\ the\ object\ and\ classify\ also$
    - $objech\ Detection: Extract\ the\ location\ also$
- $Object\ detection\ is\ used\ to\ extract\ the\ coordinates\ of\ object\ is\ called\ as\ Bounding\ box$
- $Bounding\ box\ means\ a\ rectangular\ box\ on\ the\ images\ , rectangular\ box\ will\ give$ $(x, y, x + w, y + h)$

  $top\ left\ corner = (x, y) \quad bottom\ right = (x + w, y + h)$

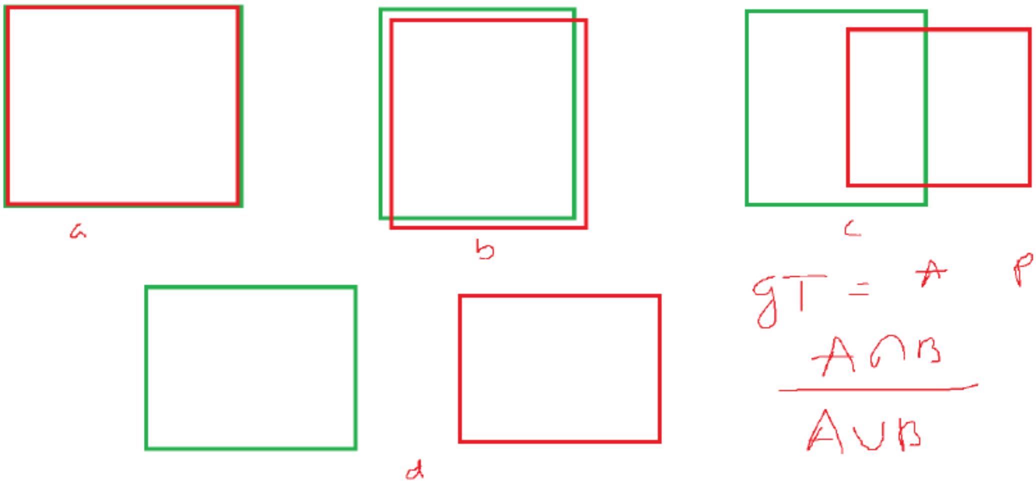


*Inter section of Union (IOU)*




*Actual boxes : Ground truth*

*Ground truth bbox vs Prediction bbox*




$gT = A \quad P = B$

$$\frac{A \cap B}{A \cup B}$$

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

*Fine Tuing*

*Consider MobileNet trained on* 1.3*Million images*

- *Identify the objects around* 1000 *classes*
- *Imagine that you want idenitfy your college ID card*
- *This model is not trained on your clg id card*
- *so It will not able to identify the clg id card*
- *You need to collect some images of your clg id card*
- *You need to create a dataset : around* 100 *images*
- *Again you need to train the MobileNet model*
- *Mobile net already has* 1000 *classes* + 1 *classes*
- *according to this agian weights wil be updated*
- *Fine Tuing*